

Day 14 - Peer Consolidation

My EDA Findings



No missing values, duplicate records, or invalid bit labels were found



The dataset shows a moderate class imbalance (60.65% class 0, 39.35% class 1).



Analog Value and Analog Voltage are the most important features for bit classification.



NTU has a noticeable impact on signal behavior and classification performance.



Analog Value and Analog Voltage are highly correlated, indicating feature redundancy.

Difficult Classes / Conditions:

Bit classification becomes more challenging at higher turbidity (NTU) levels due to increased signal degradation.

Peer Insight :

- Comparing performance across individual NTU levels may reveal how turbidity affects classification accuracy and BER.

Hypothesis :

- As NTU increases, the received optical signal becomes more distorted, causing fixed-threshold detection performance to decrease, while machine learning models can better adapt to changing signal patterns and maintain higher classification accuracy.

